

CASE: Solar Monitoring Platform

A cloud-based system that monitors and analyzes data from solar panels, inverters, and modules in real-time to optimize performance and detect faults.

Section1: SRS (Software Requirement Specification)

Problem Domain

The renewable energy sector faces major issues in monitoring solar installations:

- Lack of real-time visibility into panel performance
- Difficulty in detecting faulty panels or inverters
- Inefficient maintenance scheduling
- Data scattered across multiple devices (IoT fragmentation)

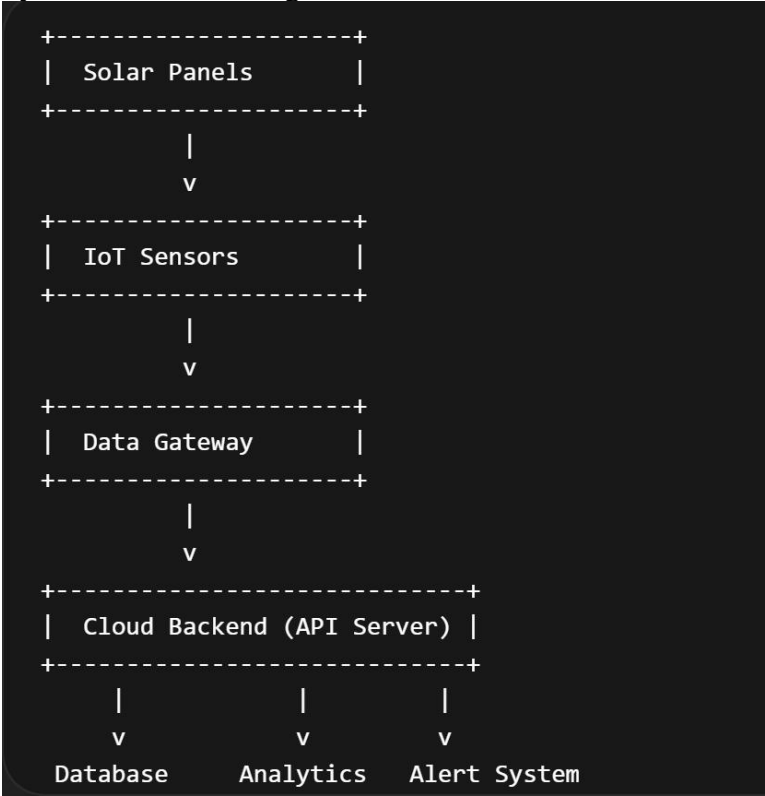
Example: A solar farm may lose 15–20% efficiency due to undetected faults.

Solution Domain

A Solar Monitoring Platform that:

- Collects real-time data from solar devices (IoT sensors)
- Displays analytics via dashboards
- Sends alerts for faults
- Uses AI/ML for predictive maintenance

System Overview Diagram



Stakeholders

- Solar farm owners
- Maintenance teams
- Energy companies
- Government agencies

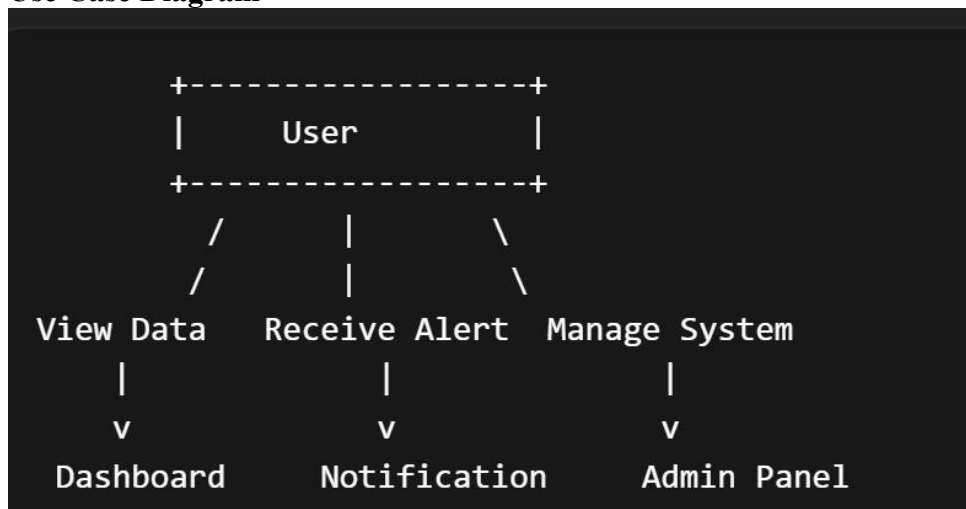
Functional Requirements

1. Real-time monitoring of:
 - a) Panel voltage/current
 - b) Inverter status
 - c) Energy generation
2. Fault Detection:
 - a) Identify underperforming panels
 - b) Detect inverter failures
3. Alerts:
 - a) SMS/Email notifications
4. Dashboard:
 - a) Graphs of energy production
 - b) Historical data
5. User Management:
 - a) Admin / Operator roles

Non-Functional Requirements

- High availability (99.9%)
- Scalable (support 10,000+ devices)
- Secure (data encryption, authentication)
- Low latency (<2 seconds for alerts)

Use Case Diagram



Section 2: IPR (Intellectual Property Rights)

IPR ensures your software is legally protected from copying or misuse.

How This Software is Protected

1. Trade Secrets
 - Data processing algorithms
 - Fault detection logic
 - ML models
 - These are kept confidential (not publicly disclosed)
2. Licensing
 - Software can be licensed (SaaS model)
 - Users pay subscription fees
3. Contracts & NDAs
 - Employees sign NDA
 - Prevents code/data leakage
4. Trademarks
 - Brand name (e.g., “SolarWatch AI”)
 - Logo and UI branding

Example: If a competitor copies your fault detection algorithm → you can take legal action under IPR laws.

Section 3: Copyright (Protecting Source Code)

What is Protected?

- Source code (Python, JavaScript, etc.)
- UI design
- Documentation

How Copyright Works

- Automatically granted when code is created
- No need to file (but registration strengthens case)

Protection Methods

1. Code Ownership: Company owns code (via employment agreement)
2. Version Control Evidence: GitHub/GitLab commits prove authorship
3. Copyright Notice

Example:

© 2026 Solar Monitoring Platform. All rights reserved.

Example Violation: If someone copies your dashboard UI or backend logic → it is a copyright violation.

Code Protection Architecture

```
Developer → Writes Code → Stored in Git Repo → Deployed to Cloud
          |
          v
        Copyright Protected
```

Section 4- Patents (Protecting Software Methods)

What Can Be Patented?

Software itself is hard to patent, but you can patent:

- Unique algorithms
- Data processing techniques
- IoT communication methods

Possible Patents

- 1. Fault Detection Algorithm:** AI model predicting panel failure before it happens
- 2. Energy Optimization Logic:** Automatically redistributing load across panels
- 3. Smart IoT Data Compression:** Reduces bandwidth while sending sensor data

Patent Flow Diagram

Idea → Prototype → Patent Filing → Examination → Approval → Protection (20 years)

Example Patent Claim

"A system that predicts solar panel degradation using real-time IoT data and machine learning."

Why Patent Matters

Without a patent:

- Competitors can copy your innovation

With a patent:

- You get exclusive rights

Conclusion

The Solar Monitoring Platform:

- Solves real-world renewable energy problems
- Uses IoT + Cloud + AI
- Can be protected through:
 - IPR (overall protection)
 - Copyright (code)
 - Patents (algorithms & methods)